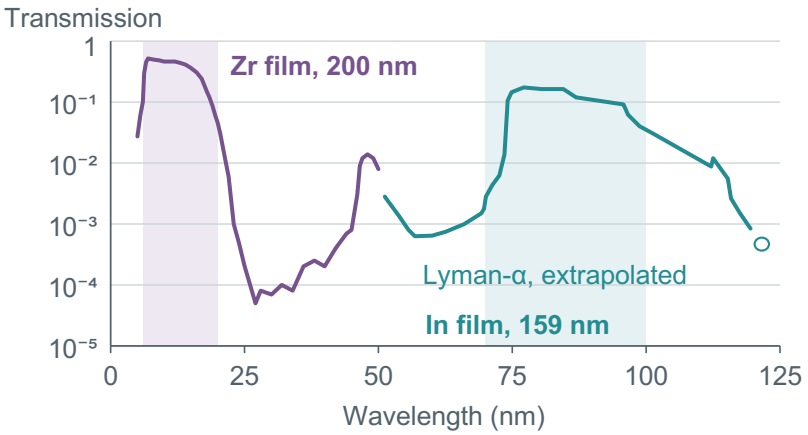
Component evidence and the calibration plan

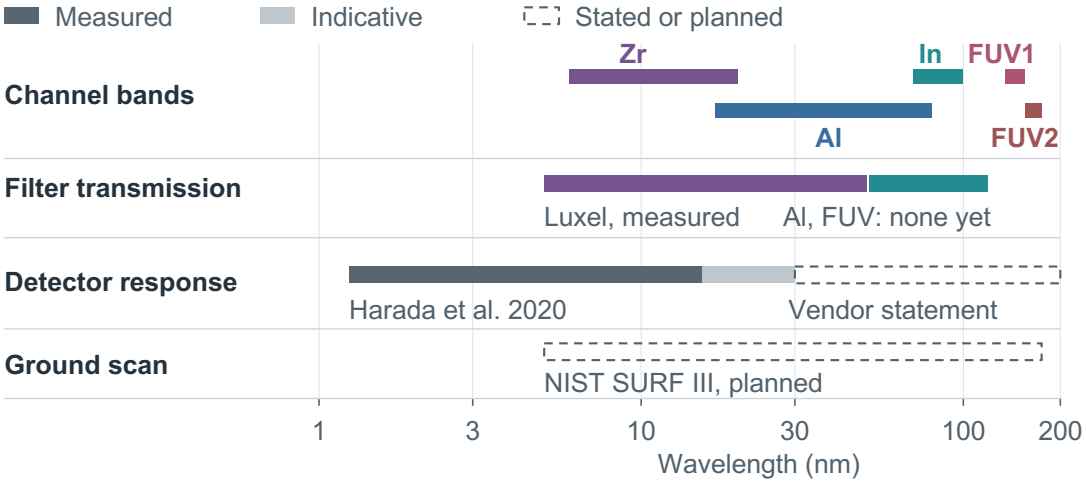
Measured component data, calibration plan, and bench results

a Measured filter transmission



Shading: nominal channel bands. Data: Luxel Corp.

b Where measured evidence exists



c Calibration plan

1 Every occultation

Reference rows above the atmosphere give I_0.
Stable common gain and brightness cancel in I/I_0.
Flat-field drift: a 1 % random term. Flares reject a pass.

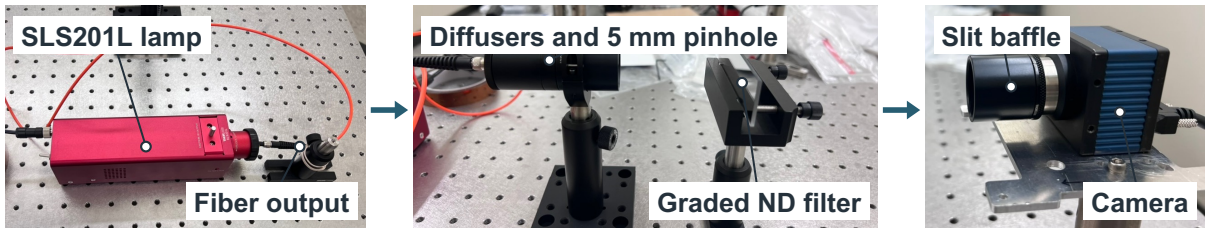
2 Every occultation

Frames below 105 km, where EUV channels see no in-band light, measure the out-of-band leak so it can be subtracted.

3 On the ground, before launch

NIST SURF III, 5–175 nm, measures the engineering unit's response and a witness filter's optical depth; the flight films share its Luxel run.

d Visible-light bench test, Carthage



Flight and bench optics

	Flight	Bench
Source angle	9.3 mrad	10.6 mrad
Slit width	25 µm	100 µm
Slit to detector	5.0 mm	17.5 mm
Blur ÷ slit width	**1.86**	**1.86**
Fresnel number	**0.71–20.8**	**0.74–1.02**
FUV Fresnel	**0.71–0.93**	**0.74–1.02**
Wavelength	6–175 nm	580–800 nm

Bench tests

1 **Optical depth from I/I_0**
Measured $|\Delta\tau| \leq 0.0096$ for $\tau = 0.6–2.25$.

2 **Image position over 10 min**
Position drift ≤0.05 detector row.

3 **Two absorbers at once**
Recover mixture from separate responses.

Figure 3. (a) Measured Zr and In transmission; the Lyman-α point is extrapolated. (b) Evidence coverage. (c) In-flight corrections and planned synchrotron calibration. (d) The visible-light bench test matches the blur-to-slit ratio and overlaps the FUV diffraction regime; it does not reproduce the full EUV range. Test 1 is measured against a 0.02 limit; tests 2 and 3 are targets.